Mania Sultana

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PROFESSIONAL SUMMARY

Research-oriented Computer Science & Engineering graduate specializing in Artificial Intelligence, Machine Learning, Robotics, Computer Vision, Human-Computer Interaction, and Intelligent Software Systems. Experienced in research, AI-driven robotics, enterprise software development, and scientific publication. 2nd author of a Springer conference paper ICTIS 2025 International Conference, held at Cornell University, New York, USA.

RESEARCH & PROFESSIONAL EXPERIENCE

Research Assistant | Institute for Advanced Research, UIU Jan to jun 2025

- Worked on ML-Based Cross-Tier Data Security for WBANs research project.
- Assisted in machine learning implementation, research analysis, and publication preparation.
- Conducted intelligent healthcare security research and technical documentation.

TECHNICAL SKILLS

AI & Robotics: YOLOv8, OpenCV, Arduino, Computer Vision, Robotics, Sensor Integration, Servo Motor Control
Programming: Python, PHP.
Web Technologies: HTML5, CSS3, Bootstrap.
Database & Tools: VS Code, XAMPP, Overleaf, Colab, MATLAB
Research Areas: Artificial Intelligence, Machine Learning, Intelligent Systems, Data-efficient, Software Engineering, Robotics, Human-Computer Interaction (HCI), Human-Robot Interaction (HRI)

FEATURED PROJECTS

AI-Based Recycling & Non-Recycling Automated Robotic Arm

- Developed an AI-powered robotic arm using YOLOv8.
- Automated recyclable and non-recyclable waste sorting and Integrated Arduino Uno, servo motors, and computer vision.

AI-Based Student Clustering System | Machine Learning Project

- Developed intelligent student grouping and section allocation system.
- Applied K-Means, Hierarchical Clustering, and Jaccard Similarity techniques.

Other Projects

- Event Management System
- Student Loan & Crowdfunding Platform
- Hostel Management System
- Smart City Management System

PUBLICATIONS & PRESENTATIONS

ICTIS 2025 International Conference Cornell University, USA
Paper: *Energy Efficient Software Development Through Algorithm Optimization*

- 2nd Author of Springer conference proceedings publication.
- Presented research on intelligent software optimization techniques.

EDUCATION

Bachelor of Science in Computer Science & Engineering
United International University (UIU), Bangladesh Spring 2026
Major: Software Engineering

ACHIEVEMENTS

- Research Assistant
- ICTIS 2025 International Conference Presenter
- Springer Conference Publication (2nd Author)
- Research Assistant under UIU Research Grant
- 2nd Runner-Up Award Winner
- AI Robotics Project Developer
- Computer Vision Research